



PARIS PANTHÉON-ASSAS UNIVERSITÉ

Marceau Bilger

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Database Project : Artconnect Application

Step 3

1) Create and populate the database

Prompt for data insertion :

“ I have the following SQL statement to create a database for an application about artists and exhibits :

```sql script for database creation```

Create a sample data insertion script (artists, works, events, members...) to demonstrate interesting cases (multiple artists, multiple events, cross-participations, etc.). ”

### OUTPUT :

```
-- =====
-- SAMPLE DATA INSERTION SCRIPT
-- Artists & Exhibitions Database
-- Demonstrates: multiple artists, cross-discipline works,
-- shared exhibitions, multi-instructor workshops,
-- community reviews and bookings.
-- =====

-- -----
-- 1. DISCIPLINES
-- -----
INSERT INTO Discipline (discName) VALUES
('Painting'),
('Sculpture'),
('Photography'),
('Digital Art'),
('Ceramics'),
('Printmaking'),
('Installation Art');

-- -----
-- 2. ARTWORK TAGS
-- -----
INSERT INTO ArtworkTag (name) VALUES
('abstract'),
('portrait'),
('landscape'),
('political'),
('minimalist'),
('surreal'),
('figurative'),
('experimental'),
('urban'),
```

('nature');

---

-- 3. COMMUNITY MEMBERS

---

INSERT INTO CommunityMember (ComID, Name, email, birthyear, phone, city, membershipType) VALUES

(1, 'Alice Moreau', 'alice.moreau@email.com', 1985, '0612345678', 'Paris', 'Premium'),  
(2, 'Benoît Lefèvre', 'benoit.lefevre@email.com', 1990, '0623456789', 'Lyon', 'Standard'),  
(3, 'Camille Fontaine', 'camille.fontaine@email.com', 1978, '0634567890', 'Bordeaux', 'Premium'),  
(4, 'David Nguyen', 'david.nguyen@email.com', 2000, '0645678901', 'Paris', 'Student'),  
(5, 'Émilie Rousseau', 'emilie.rousseau@email.com', 1995, '0656789012', 'Marseille', 'Standard'),  
(6, 'François Bernard', 'francois.bernard@email.com', 1968, '0667890123', 'Toulouse', 'Premium'),  
(7, 'Gina Kowalski', 'gina.kowalski@email.com', 2001, '0678901234', 'Paris', 'Student'),  
(8, 'Henri Dubois', 'henri.dubois@email.com', 1975, '0689012345', 'Nantes', 'Standard');

---

-- 4. FAVORITE DISCIPLINES (members with multiple interests)

---

INSERT INTO FavoriteDiscipline (ComID, discName) VALUES

(1, 'Painting'), (1, 'Photography'),  
(2, 'Sculpture'), (2, 'Installation Art'),  
(3, 'Photography'), (3, 'Digital Art'),  
(4, 'Digital Art'), (4, 'Printmaking'),  
(5, 'Ceramics'), (5, 'Painting'),  
(6, 'Sculpture'), (6, 'Painting'),  
(7, 'Photography'), -- note: reuses tag as interest  
(8, 'Installation Art');

-- (Fix: FavoriteDiscipline references Discipline, so remove invalid 'Surreal')

-- DELETE FROM FavoriteDiscipline WHERE ComID = 7 AND discName = 'Surreal';

-- INSERT INTO FavoriteDiscipline (ComID, discName) VALUES (7, 'Photography');

---

-- 5. GALLERIES

---

INSERT INTO Gallery (GalleryID, name, address, ownerName, openingHours, contactPhone, rating, website) VALUES

(1, 'Galerie Lumière', '12 Rue du Faubourg, Paris', 'Marie Clair', 'Tue-Sat 10:00-19:00', '0142345678', 4.7, 'www.galerie-lumiere.fr'),  
(2, 'Espace Contemporain', '45 Quai des Arts, Lyon', 'Pierre Aubert', 'Wed-Sun 11:00-18:00', '0478901234', 4.2, 'www.espace-contemporain.fr'),

(3, 'Atelier du Sud', '8 Boulevard du Mistral, Marseille', 'Sophie Blanc', 'Mon-Sat 09:00-17:00', '0491234567', 4.5, 'www.atelier-du-sud.fr');

-- 6. ARTISTS (6 artists, some multi-discipline, some active/inactive)

INSERT INTO Artist (artistID, name, birthYear, contactEmail, phone, city, website, SocialMedia, isActive) VALUES

(1, 'Léa Tremblay', 1980, 'lea.tremblay@art.com', '0611111111', 'Paris', 'www.leadtremblay.com', '@leadtremblay', TRUE),  
(2, 'Marco Ferretti', 1975, 'marco.ferretti@art.com', '0622222222', 'Marseille', 'www.marcoferretti.it', '@marcoferretti', TRUE),  
(3, 'Naomi Okoro', 1990, 'naomi.okoro@art.com', '0633333333', 'Lyon', 'www.naomiokoro.com', '@naomiokoro', TRUE),  
(4, 'Samuel Vidal', 1968, 'samuel.vidal@art.com', '0644444444', 'Bordeaux', 'www.samuelvidal.fr', '@samuelvidal', FALSE), -- retired  
(5, 'Yuki Hayashi', 1995, 'yuki.hayashi@art.com', '0655555555', 'Paris', 'www.yukihayashi.jp', '@yukihayashi', TRUE),  
(6, 'Clara Martínez', 1985, 'clara.martinez@art.com', '0666666666', 'Toulouse', 'www.claramartinez.es', '@claramartinez', TRUE);

-- 7. ARTIST DISCIPLINES (cross-discipline artists)

INSERT INTO Disciplines (discName, artistID) VALUES

('Painting', 1), -- Léa: Painting + Photography  
( 'Photography', 1),  
( 'Sculpture', 2), -- Marco: Sculpture + Installation Art  
( 'Installation Art', 2),  
( 'Photography', 3), -- Naomi: Photography + Digital Art  
( 'Digital Art', 3),  
( 'Painting', 4), -- Samuel: Painting + Printmaking (retired)  
( 'Printmaking', 4),  
( 'Digital Art', 5), -- Yuki: Digital Art + Installation Art  
( 'Installation Art', 5),  
( 'Ceramics', 6), -- Clara: Ceramics + Sculpture  
( 'Sculpture', 6);

-- 8. ARTWORKS

INSERT INTO Artwork (artworkID, title, creationYear, type, medium, dimensions, price, description, artistID) VALUES

-- Léa Tremblay (Painting + Photography)

(1, 'Reflets Urbains', 2021, 'Painting', 'Oil on canvas', '120x90cm', 3500.00, 'City reflections in rain puddles', 1),  
(2, 'Mémoire Floue', 2022, 'Photography', 'Fine art print', '80x60cm', 1200.00, 'Blurred long-exposure urban portrait', 1),

(3, 'Lumière d"Automne', 2023, 'Painting', 'Acrylic on canvas', '100x80cm', 4200.00, 'Abstract autumn light study', 1),

-- Marco Ferretti (Sculpture + Installation)

(4, 'Équilibre Précaire', 2019, 'Sculpture', 'Welded steel', '45x30x30cm', 8000.00, 'Balanced steel forms exploring tension', 2),

(5, 'Void IV', 2022, 'Installation', 'Mixed media, mirrors', '300x200cm', 15000.00, 'Room-scale mirror installation', 2),

-- Naomi Okoro (Photography + Digital Art)

(6, 'Identités', 2020, 'Photography', 'Archival pigment print', '50x70cm', 2200.00, 'Portrait series on cultural identity', 3),

(7, 'Glitch Portraits', 2023, 'Digital Art', 'Digital print on alu', '60x90cm', 1800.00, 'AI-assisted glitched portrait series', 3),

(8, 'Data Drift', 2022, 'Digital Art', 'Generative, 3-screen', '200x80cm', 9500.00, 'Generative data-driven landscape', 3),

-- Samuel Vidal (retired — historic works)

(9, 'Géographies I', 2005, 'Painting', 'Oil on linen', '180x150cm', 6500.00, 'Early geometric landscape series', 4),

(10, 'Empreintes', 2010, 'Printmaking', 'Etching, ed. 30', '40x50cm', 850.00, 'Series of architectural etchings', 4),

-- Yuki Hayashi (Digital + Installation)

(11, 'Pixel Garden', 2022, 'Digital Art', 'Interactive screen', '120x80cm', 5500.00, 'Interactive generative garden', 5),

(12, 'Floating Grid', 2023, 'Installation', 'Neon tubes, sensors', '400x300cm', 22000.00, 'Sensor-reactive neon installation', 5),

-- Clara Martínez (Ceramics + Sculpture)

(13, 'Terre et Feu I', 2021, 'Ceramics', 'Stoneware, raku-fired', '30x20x20cm', 650.00, 'Raku-fired vessel exploring fire marks', 6),

(14, 'Corps Fragmentés', 2023, 'Sculpture', 'Terracotta', '60x40x40cm', 3200.00, 'Fragmented human form in terracotta', 6),

(15, 'Argile Vivante', 2023, 'Ceramics', 'Porcelain, hand-built', '25x15x15cm', 480.00, 'Organic hand-built porcelain forms', 6);

-----  
-- 9. ARTWORK TAGS (artworks with multiple tags)  
-----

INSERT INTO Tags (artworkID, name) VALUES

(1, 'urban'), (1, 'abstract'),

(2, 'portrait'), (2, 'urban'),

(3, 'abstract'), (3, 'landscape'),

(4, 'minimalist'),

(5, 'experimental'), (5, 'abstract'),

(6, 'portrait'), (6, 'political'),

(7, 'portrait'), (7, 'experimental'),

(8, 'abstract'), (8, 'landscape'), (8, 'experimental'),  
(9, 'landscape'), (9, 'abstract'),  
(10, 'figurative'),  
(11, 'abstract'), (11, 'nature'), (11, 'experimental'),  
(12, 'minimalist'), (12, 'experimental'),  
(13, 'abstract'), (13, 'nature'),  
(14, 'figurative'),  
(15, 'nature'), (15, 'abstract');

-- -----  
-- 10. EXHIBITIONS (cross-gallery, mixed themes)  
-- -----

INSERT INTO Exhibition (exhibitionID, title, startDate, endDate, description, curatorName, theme, GalleryID) VALUES

(1, 'Fragments du Réel', '2023-03-01', '2023-04-30', 'Exploring fragmented urban realities', 'Claire Vernet', 'Urban & Identity', 1),  
(2, 'Matière et Mémoire', '2023-06-15', '2023-08-31', 'Material memory across generations', 'Paul Desroches', 'Heritage & Memory', 2),  
(3, 'Nouvelles Géographies', '2023-09-01', '2023-11-30', 'Digital and physical landscape exploration', 'Irène Morel', 'Landscape & Space', 1),  
(4, 'Corps & Territoire', '2024-01-20', '2024-03-31', 'Body as landscape and political territory', 'Claire Vernet', 'Body & Politics', 3),  
(5, 'Biennale Jeune Création', '2024-05-01', '2024-07-31', 'Emerging artists under 35', 'Alain Petit', 'Emerging Voices', 2);

-- -----  
-- 11. EXHIBITS (artworks in exhibitions — cross-participations)  
-- -----

-- Exhibition 1: Fragments du Réel (Paris)

INSERT INTO exhibits (artworkID, exhibitionID) VALUES

(1, 1), -- Léa: Reflets Urbains  
(2, 1), -- Léa: Mémoire Floue  
(6, 1), -- Naomi: Identités  
(4, 1), -- Marco: Équilibre Précaire  
(14, 1); -- Clara: Corps Fragmentés

-- Exhibition 2: Matière et Mémoire (Lyon)

INSERT INTO exhibits (artworkID, exhibitionID) VALUES

(9, 2), -- Samuel: Géographies I (retired master)  
(10, 2), -- Samuel: Empreintes  
(13, 2), -- Clara: Terre et Feu I  
(4, 2), -- Marco: Équilibre Précaire ← same artwork, 2 shows  
(1, 2); -- Léa: Reflets Urbains ← same artwork, 2 shows

-- Exhibition 3: Nouvelles Géographies (Paris)

INSERT INTO exhibits (artworkID, exhibitionID) VALUES

(3, 3), -- Léa: Lumière d'Automne  
(8, 3), -- Naomi: Data Drift

(11, 3), -- Yuki: Pixel Garden  
(12, 3), -- Yuki: Floating Grid  
(9, 3); -- Samuel: Géographies I ← 3rd show for this historic piece

-- Exhibition 4: Corps & Territoire (Marseille)

INSERT INTO exhibits (artworkID, exhibitionID) VALUES

(6, 4), -- Naomi: Identités ← 2nd show  
(7, 4), -- Naomi: Glitch Portraits  
(14, 4), -- Clara: Corps Fragmentés ← 2nd show  
(5, 4), -- Marco: Void IV  
(2, 4); -- Léa: Mémoire Floue ← 2nd show

-- Exhibition 5: Biennale Jeune Création (Lyon) — artists under 35

INSERT INTO exhibits (artworkID, exhibitionID) VALUES

(7, 5), -- Naomi: Glitch Portraits ← 2nd show  
(11, 5), -- Yuki: Pixel Garden ← 2nd show  
(12, 5), -- Yuki: Floating Grid ← 2nd show  
(15, 5); -- Clara: Argile Vivante

-----  
-- 12. WORKSHOPS  
-----

INSERT INTO Workshop (workshopID, title, workDate, durationMinutes, maxParticipants, price, location, description, level) VALUES

(1, 'Introduction à l'aquarelle', '2023-10-07', 180, 12, 45.00, 'Galerie Lumière, Paris', 'Watercolour basics for beginners', 'Beginner'),  
(2, 'Photographie de rue', '2023-10-14', 240, 8, 80.00, 'Paris — Marais district', 'Street photography guided walk', 'Intermediate'),  
(3, 'Initiation à la céramique', '2023-11-04', 300, 10, 65.00, 'Atelier du Sud, Marseille', 'Hand-building and basic glazing', 'Beginner'),  
(4, 'Art numérique et IA', '2023-11-18', 240, 15, 90.00, 'Espace Contemporain, Lyon', 'Generative art with AI tools', 'Intermediate'),  
(5, 'Sculpture sur métal', '2024-01-13', 360, 6, 120.00, 'Atelier du Sud, Marseille', 'Steel welding and form exploration', 'Advanced'),  
(6, 'Installation in situ', '2024-02-24', 480, 8, 150.00, 'Espace Contemporain, Lyon', 'Creating site-specific installation works', 'Advanced'),  
(7, 'Portrait photographique', '2024-03-09', 180, 10, 70.00, 'Galerie Lumière, Paris', 'Studio portrait lighting and posing', 'Beginner');

-----  
-- 13. INSTRUCTORS (some workshops have 2 instructors)  
-----

INSERT INTO Instructor (workshopID, artistID) VALUES

(1, 1), -- Léa teaches watercolour  
(2, 1), -- Léa teaches street photography  
(2, 3), -- Naomi co-teaches street photography ← dual instructor  
(3, 6), -- Clara teaches ceramics  
(4, 3), -- Naomi teaches digital art

(4, 5), -- Yuki co-teaches digital art ← dual instructor  
 (5, 2), -- Marco teaches metal sculpture  
 (6, 2), -- Marco teaches installation  
 (6, 5), -- Yuki co-teaches installation ← dual instructor  
 (7, 1), -- Léa teaches portrait photography  
 (7, 3); -- Naomi co-teaches portrait photography ← triple instructor occurrence for Naomi

---

## -- 14. BOOKINGS

---

INSERT INTO Booking (BookingID, BookingDate, paymentStatus, workshopID, ComID)  
 VALUES

-- Workshop 1 (watercolour) — 5 members

(1, '2023-09-20', 'Paid', 1, 1),  
 (2, '2023-09-22', 'Paid', 1, 5),  
 (3, '2023-09-25', 'Paid', 1, 7),  
 (4, '2023-10-01', 'Pending', 1, 8),  
 (5, '2023-10-03', 'Paid', 1, 2),

-- Workshop 2 (street photography) — Alice and David both attend

(6, '2023-09-28', 'Paid', 2, 1), -- Alice: watercolour AND photography  
 (7, '2023-09-30', 'Paid', 2, 4),  
 (8, '2023-10-02', 'Paid', 2, 3),

-- Workshop 3 (ceramics) — Émilie attends in her favourite discipline

(9, '2023-10-15', 'Paid', 3, 5),  
 (10, '2023-10-18', 'Paid', 3, 6),  
 (11, '2023-10-20', 'Cancelled', 3, 8), -- Henri cancelled

-- Workshop 4 (digital art) — tech-savvy members

(12, '2023-11-01', 'Paid', 4, 3),  
 (13, '2023-11-02', 'Paid', 4, 4), -- David: photography AND digital art  
 (14, '2023-11-05', 'Paid', 4, 7),

-- Workshop 5 (metal sculpture — advanced, expensive)

(15, '2024-01-03', 'Paid', 5, 2), -- Benoît: loves sculpture  
 (16, '2024-01-05', 'Paid', 5, 6),

-- Workshop 6 (installation — advanced)

(17, '2024-02-01', 'Paid', 6, 2), -- Benoît: sculpture AND installation workshops  
 (18, '2024-02-03', 'Pending', 6, 3),

-- Workshop 7 (portrait photography)

(19, '2024-02-20', 'Paid', 7, 1), -- Alice: 3 workshops total  
 (20, '2024-02-21', 'Paid', 7, 7),  
 (21, '2024-02-22', 'Paid', 7, 4);

---



-- 15. REVIEWS (members reviewing artworks they've seen)

-----  
INSERT INTO Review (reviewID, rating, comment, reviewDate, artworkID, ComID) VALUES

-- Reviews of Léa's works

(1, 5, 'Stunning play of light — the rain reflections feel alive', '2023-04-10', 1, 1),

(2, 4, 'Emotionally resonant long exposure, beautifully printed', '2023-04-25', 2, 3),

(3, 5, 'Best piece in the show — colour palette is extraordinary', '2023-10-15', 3, 6),

-- Reviews of Marco's works

(4, 4, 'Loved the tension in the sculpture, impressive balance', '2023-04-12', 4, 2),

(5, 5, 'Void IV is overwhelming in the best way — disorienting and beautiful', '2023-11-10', 5, 1),

(6, 3, 'Interesting concept but the mirrors need better maintenance', '2023-11-12', 5, 8),

-- Reviews of Naomi's works

(7, 5, 'Identités is the most politically powerful series I have seen this year', '2023-04-20', 6, 1),

(8, 4, 'Glitch Portraits: digital fragmentation mirrors real identity questions', '2024-04-15', 7, 4),

(9, 5, 'Data Drift is hypnotic — I stood in front of it for 20 minutes', '2023-10-18', 8, 5),

-- Reviews of Samuel's historic works

(10, 5, 'Géographies I remains a masterpiece — timeless geometric precision', '2023-08-10', 9, 6),

(11, 4, 'Empreintes: the etchings have a beautiful tactile quality even as prints', '2023-08-12', 10, 3),

-- Reviews of Yuki's works

(12, 5, 'Pixel Garden is addictive — every interaction changes the whole composition', '2023-10-20', 11, 7),

(13, 4, 'Floating Grid: the sensors are a bit slow but the neon effect is superb', '2023-10-25', 12, 2),

(14, 5, 'Completely immersive, the best installation of the biennale', '2024-06-10', 12, 4),

-- Reviews of Clara's works

(15, 5, 'Terre et Feu I — the raku marks are unpredictable and gorgeous', '2023-08-15', 13, 5),

(16, 4, 'Corps Fragmentés speaks volumes about body politics, powerful work', '2024-03-10', 14, 1),

(17, 5, 'Argile Vivante: organic forms that look grown, not made', '2024-07-05', 15, 7);

```
-- =====
-- END OF SCRIPT
-- =====
```

## Views :

### View 1 :

Objective: SECURITY — hide private contact data

Artists' phone numbers and personal emails must not be exposed to end-users or front-end APIs. This view surfaces only the information suitable for a public gallery website: name, city, disciplines, activity status, and public links. Internal queries that need contactEmail / phone must go directly to the Artist table (restricted to admin roles).

```
CREATE OR REPLACE VIEW PublicArtistProfile AS
SELECT
 a.artistID,
 a.name,
 a.city,
 a.website,
 a.SocialMedia,
 a.isActive,
 GROUP_CONCAT(d.discName ORDER BY d.discName SEPARATOR ', ') AS disciplines
FROM Artist a
LEFT JOIN Disciplines d ON a.artistID = d.artistID
GROUP BY
 a.artistID, a.name, a.city,
 a.website, a.SocialMedia, a.isActive;
```

### View 2 :

#### VIEW 2 · ExhibitionCatalogue

Objective: QUERY SIMPLIFICATION — join-heavy aggregation

A typical catalogue query requires joining Exhibition, Gallery, exhibits, Artwork, Artist, and Tags. This view pre-assembles the full catalogue entry for every artwork in every exhibition, making catalogue screens and PDF exports a simple SELECT with optional WHERE filters.

```
CREATE OR REPLACE VIEW ExhibitionCatalogue AS
SELECT
 ex.exhibitionID,
 ex.title AS exhibitionTitle,
 ex.startDate,
```

```

 ex.endDate,
 ex.theme,
 ex.curatorName,
 g.name AS galleryName,
 g.city AS galleryCity,
 a.artworkID,
 a.title AS artworkTitle,
 a.type AS artworkType,
 a.medium,
 a.dimensions,
 a.price AS artworkPrice,
 ar.name AS artistName,
 ar.city AS artistCity,
 GROUP_CONCAT(t.name ORDER BY t.name SEPARATOR ', ') AS tags
FROM Exhibition ex
JOIN Gallery g ON ex.GalleryID = g.GalleryID
JOIN exhibits e ON ex.exhibitionID = e.exhibitionID
JOIN Artwork a ON e.artworkID = a.artworkID
JOIN Artist ar ON a.artistID = ar.artistID
LEFT JOIN Tags t ON a.artworkID = t.artworkID
GROUP BY
 ex.exhibitionID, ex.title, ex.startDate, ex.endDate,
 ex.theme, ex.curatorName,
 g.name, g.city,
 a.artworkID, a.title, a.type, a.medium, a.dimensions, a.price,
 ar.name, ar.city;

```

View 3 :

VIEW 6 · (Bonus) ActiveExhibitions

Objective: SECURITY + SIMPLIFICATION — temporal filtering Many queries need "exhibitions currently open". Rather than writing WHERE startDate <= CURRENT\_DATE AND endDate >= CURRENT\_DATE everywhere (error-prone), this view encapsulates the temporal logic and can be granted to read-only API roles without exposing past/future shows.

CREATE OR REPLACE VIEW ActiveExhibitions AS  
SELECT

```

 ex.exhibitionID,
 ex.title,
 ex.startDate,
 ex.endDate,
 ex.theme,
 ex.curatorName,
 g.name AS galleryName,
 g.address AS galleryAddress,
 g.openingHours,
 g.contactPhone,
 g.website AS galleryWebsite,

```

```

COUNT(DISTINCT e.artworkID) AS artworkCount,
COUNT(DISTINCT a.artistID) AS artistCount
FROM Exhibition ex
JOIN Gallery g ON ex.GalleryID = g.GalleryID
JOIN exhibits e ON ex.exhibitionID = e.exhibitionID
JOIN Artwork a ON e.artworkID = a.artworkID
WHERE ex.startDate <= CURRENT_DATE
AND ex.endDate >= CURRENT_DATE
GROUP BY
 ex.exhibitionID, ex.title, ex.startDate, ex.endDate,
 ex.theme, ex.curatorName,
 g.name, g.address, g.openingHours, g.contactPhone, g.website;

```

#### Indexes:

```

CREATE INDEX idx_artwork_artistID ON Artwork(artistID);

CREATE INDEX idx_booking_workshopID ON Booking(workshopID);

CREATE INDEX idx_booking_comID ON Booking(ComID);

```

#### Triggers :

##### Trigger 1

Checks if a workshop is full before adding a new participant to it.

```

CREATE TRIGGER trg_check_workshop_capacity
BEFORE INSERT ON Booking
FOR EACH ROW
BEGIN
 DECLARE current_count INT;
 DECLARE max_cap INT;

 SELECT COUNT(*) INTO current_count
 FROM Booking
 WHERE workshopID = NEW.workshopID;

 SELECT maxParticipants INTO max_cap
 FROM Workshop
 WHERE workshopID = NEW.workshopID;

 IF current_count >= max_cap THEN
 SIGNAL SQLSTATE '45000'
 SET MESSAGE_TEXT = 'Workshop is fully booked. No more participants can be
added.';

```

```
 END IF;
END
```

Trigger 2 :

Checks the validity of a date when a new booking is added

```
CREATE TRIGGER trg_check_exhibition_dates_insert
BEFORE INSERT ON Exhibition
FOR EACH ROW
BEGIN
 IF NEW.endDate < NEW.startDate THEN
 SIGNAL SQLSTATE '45000'
 SET MESSAGE_TEXT = 'Exhibition end date cannot be before the start date.';
 END IF;
END$$
```

```
CREATE TRIGGER trg_check_exhibition_dates_update
BEFORE UPDATE ON Exhibition
FOR EACH ROW
BEGIN
 IF NEW.endDate < NEW.startDate THEN
 SIGNAL SQLSTATE '45000'
 SET MESSAGE_TEXT = 'Exhibition end date cannot be before the start date.';
 END IF;
END
```

Trigger 3

Prevents the user from booking a past workshop

```
CREATE TRIGGER trg_check_booking_date
BEFORE INSERT ON Booking
FOR EACH ROW
BEGIN
 DECLARE workshop_date DATE;

 SELECT workDate INTO workshop_date
 FROM Workshop
 WHERE workshopID = NEW.workshopID;

 IF workshop_date < CURDATE() THEN
 SIGNAL SQLSTATE '45000'
 SET MESSAGE_TEXT = 'Cannot book a workshop that has already taken place.';
```

```
 END IF;
END
```

### Functions and procedures:

#### Function 1

Computes the average rating of a artwork

```
CREATE FUNCTION fn_get_artwork_avg_rating(p_artworkID INT)
RETURNS DECIMAL(4,2)
DETERMINISTIC
READS SQL DATA
BEGIN
 DECLARE avg_rating DECIMAL(4,2);

 SELECT ROUND(AVG(rating), 2) INTO avg_rating
 FROM Review
 WHERE artworkID = p_artworkID;

 RETURN COALESCE(avg_rating, 0.00);
END
```

#### Function 2

Get number of confirmed participants for a workshop

```
CREATE FUNCTION fn_get_participant_count(p_workshopID INT)
RETURNS INT
READS SQL DATA
DETERMINISTIC
BEGIN
 DECLARE participantCount INT;

 SELECT COUNT(*)
 INTO participantCount
 FROM Booking
 WHERE workshopID = p_workshopID
 AND paymentStatus = 'Paid';

 RETURN participantCount;
END
```

Function 3:

Check if a member is already booked for a workshop

```
CREATE FUNCTION fn_is_member_booked(p_comID INT, p_workshopID INT)
RETURNS BOOLEAN
READS SQL DATA
DETERMINISTIC
BEGIN
 DECLARE bookingExists INT;

 SELECT COUNT(*)
 INTO bookingExists
 FROM Booking
 WHERE ComID = p_comID AND workshopID = p_workshopID;

 RETURN bookingExists > 0;
END
```

Procedure 1 :

Create a workshop and adds an instructor to it

```
CREATE PROCEDURE sp_create_workshop_with_instructor(
 IN p_workshopID INT,
 IN p_title VARCHAR(50),
 IN p_workDate DATE,
 IN p_durationMinutes INT,
 IN p_maxParticipants INT,
 IN p_price DOUBLE,
 IN p_location VARCHAR(50),
 IN p_description VARCHAR(50),
 IN p_level VARCHAR(50),
 IN p_artistID INT
)
BEGIN
 DECLARE artist_exists INT;

 SELECT COUNT(*) INTO artist_exists
 FROM Artist
 WHERE artistID = p_artistID AND isActive = TRUE;

 IF artist_exists = 0 THEN
 SIGNAL SQLSTATE '45000'
 SET MESSAGE_TEXT = 'Artist does not exist or is not active.';
 END IF;
```

START TRANSACTION;

```
INSERT INTO Workshop
(workshopID, title, workDate, durationMinutes,
maxParticipants, price, location, description, level)
VALUES
(p_workshopID, p_title, p_workDate, p_durationMinutes,
p_maxParticipants, p_price, p_location, p_description, p_level);
```

```
INSERT INTO Instructor (workshopID, artistID)
VALUES (p_workshopID, p_artistID);
```

COMMIT;

```
SELECT CONCAT('Workshop "', p_title, '" created and artist ', p_artistID, ' registered
as instructor.') AS result;
END
```

Procedure 2:

Add a new artist with their discipline(s)

```
CREATE PROCEDURE sp_add_artist_with_discipline(
 IN p_artistID INT,
 IN p_name VARCHAR(50),
 IN p_birthYear INT,
 IN p_email VARCHAR(50),
 IN p_phone VARCHAR(50),
 IN p_city VARCHAR(50),
 IN p_website VARCHAR(50),
 IN p_discipline VARCHAR(50)
)
BEGIN
 INSERT INTO Artist (artistID, name, birthYear, contactEmail, phone, city, website,
 SocialMedia, isActive)
 VALUES (p_artistID, p_name, p_birthYear, p_email, p_phone, p_city, p_website, "",
 TRUE);

 IF NOT EXISTS (SELECT 1 FROM Discipline WHERE discName = p_discipline) THEN
 INSERT INTO Discipline (discName) VALUES (p_discipline);
 END IF;

 INSERT INTO Disciplines (discName, artistID)
 VALUES (p_discipline, p_artistID);
END
```

Procedure 3:



Get full workshop report

```
CREATE PROCEDURE sp_workshop_report(IN p_workshopID INT)
BEGIN
 -- Workshop details
 SELECT
 w.workshopID,
 w.title,
 w.workDate,
 w.level,
 w.price,
 w.maxParticipants,
 fn_get_participant_count(w.workshopID) AS paidParticipants,
 (w.maxParticipants - fn_get_participant_count(w.workshopID)) AS spotsLeft
 FROM Workshop w
 WHERE w.workshopID = p_workshopID;

 -- Instructors
 SELECT a.name AS instructorName, a.city
 FROM Instructor i
 JOIN Artist a ON i.artistID = a.artistID
 WHERE i.workshopID = p_workshopID;

 -- Registered members
 SELECT cm.Name AS memberName, cm.membershipType, b.paymentStatus,
 b.BookingDate
 FROM Booking b
 JOIN CommunityMember cm ON b.ComID = cm.ComID
 WHERE b.workshopID = p_workshopID
 ORDER BY b.BookingDate;
END
```

### Transaction :

Register a member for multiple workshops atomically

```
START TRANSACTION;

-- Attempt to book member 5 (Emma) into workshops 1 and 3
INSERT INTO Booking (BookingID, BookingDate, paymentStatus, workshopID, ComID)
VALUES (16, '2024-06-20', 'Paid', 1, 5);

INSERT INTO Booking (BookingID, BookingDate, paymentStatus, workshopID, ComID)
VALUES (17, '2024-06-20', 'Paid', 3, 5);

COMMIT;
```